Solvency Field Theory: Repaired Action, Exact Exterior Closure, and Modified Interior TOV Branch

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Status: live repaired action candidate

Abstract

This document states the current live action candidate for Solvency Field Theory after the action program's hard gates and subsequent repair sequence. The absolute exponential potential fails because it obstructs exact static vacuum. The repaired action replaces the absolute potential with a quadratic background-relative lag potential and thereby admits an exact background-matched static vacuum branch on which every non-Einstein sector vanishes. The action then reduces exactly to Einstein-Hilbert in the exterior, so the static, asymptotically flat vacuum equation is exactly $G_{\mu\nu} = 0$ and Schwarzschild follows. The repaired action also preserves the dust, radiation, and vacuum minisuperspace branches, and yields a structured modified Tolman-Oppenheimer-Volkoff branch in matter. The exact branch mass equation, the cleaned-up modified radial pressure, center regularity, and the weak-field/Newtonian limit are stated explicitly. What remains open is narrow: the final covariant/background-independent packaging of the cosmological clock branch, a full perfect-fluid variational closure of the radial sector, and deeper interior phenomenology.

1. Purpose and status

This is the main action document, not the full derivation archive. Its purpose is to state the live repaired action, record what is now exact, and separate exact results from branch-level effective-fluid closures.

The action is no longer in the purely conjectural stage. The following points are now established at the action level or the repaired branch level:

- the source sector has a clean covariant structure and its hidden variation is controlled by clock-frame momentum flux,
- the minimal count sector has survived the first local-action gates,
- the kinetic regulator can be set to $\varepsilon = 0$ in the surviving branch without breaking the core minisuperspace structure,
- the absolute potential $V_0 e^{-\chi}$ is not the correct potential because it obstructs exact static vacuum,
- the repaired quadratic background-relative potential restores exact static vacuum exterior collapse,
- the repaired action preserves the dust, radiation, and vacuum minisuperspace branches,

- the repaired static interior branch yields an exact mass equation and a cleaned-up modified TOV skeleton.

So the right label is: *live repaired action candidate with exact exterior closure and derived interior branch equations.*

2. Live repaired action

We package the single bookkeeping scalar as

$$\chi(x) = \bar{\chi}(x) + \psi(x), \tag{1}$$

where $\bar{\chi}(x)$ is the cosmological/background bookkeeping branch and $\psi(x)$ is the local lag relative to that branch.

The effective covariant action we now carry forward is

$$\boxed{S_{\rm SFT}^{\rm eff} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\,\nabla_\mu(\bar{\chi}+\psi)\,J^\mu_{\rm ops} + \Lambda_C\,\mathcal{C}_{\rm min} - V_{\rm quad} + \mathcal{L}_m\right].} \tag{2}$$

The clock congruence is defined by the total clock field,

$$u^\mu = \frac{\nabla^\mu(\bar{\chi}+\psi)}{\sqrt{-\nabla_\alpha(\bar{\chi}+\psi)\nabla^\alpha(\bar{\chi}+\psi)}}. \tag{3}$$

The spatial projector is

$$h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu. \tag{4}$$

The covariant operation density and current are

$$\rho_{\rm ops} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h}, \qquad J^\mu_{\rm ops} = \rho_{\rm ops}u^\mu. \tag{5}$$

The minimal count sector is

$$\boxed{\mathcal{C}_{\rm min} = \mathcal{H}_\chi^2 - \alpha(E_\chi + V_{\rm quad}), \qquad \mathcal{H}_\chi = \frac{1}{3}\nabla_\mu u^\mu, \qquad E_\chi = T_{\mu\nu}u^\mu u^\nu,} \tag{6}$$

with

$$\alpha = \frac{8\pi G}{3}\,\beta, \qquad \beta = \frac{1}{4\pi^2}. \tag{7}$$

The repaired potential is

$$\boxed{V_{\rm quad} = V_0\left(e^{-\psi} - 1\right)^2.} \tag{8}$$

3. Why the repair was necessary

The original absolute potential

$$V_{\rm abs} = V_0 e^{-\chi} \tag{9}$$

obstructs exact static vacuum. On a static Killing-aligned clock branch with

$$\chi = \chi_0 + \kappa t, \tag{10}$$

one finds

$$\theta = \nabla_\mu u^\mu = 0, \qquad \mathcal{H}_\chi = 0, \qquad J^\mu_{\rm ops} = 0, \qquad E_\chi = 0, \tag{11}$$

but the potential survives as

$$V_{\rm abs} = V_0 e^{-\chi_0 - \kappa t}, \tag{12}$$

so the vacuum count sector becomes

$$\mathcal{C}^{\rm vac}_{\rm min} = -\alpha V_{\rm abs}, \tag{13}$$

and the reduced χ-equation still carries a source proportional to $V_{\rm abs}$. So the absolute potential is not compatible with exact static vacuum.

A simple relative potential,

$$V_{\rm rel} = V_0 e^{-(\chi - \bar{\chi})}, \tag{14}$$

removes the explicit time dependence but still leaves a nonzero vacuum load at $\psi = 0$. A shifted relative potential,

$$V_{\rm shift} = V_0 \left(e^{-(\chi - \bar{\chi})} - 1 \right), \tag{15}$$

fixes the vacuum value but not the first derivative. The repaired quadratic potential is the first candidate that satisfies both

$$V_{\rm quad}\big|_{\psi=0} = 0, \qquad \frac{dV_{\rm quad}}{d\psi}\Big|_{\psi=0} = 0, \tag{16}$$

while also preserving a positive restoring curvature,

$$\frac{d^2 V_{\rm quad}}{d\psi^2}\Big|_{\psi=0} = 2V_0 > 0. \tag{17}$$

4. Exact static vacuum exterior closure

On the exact background-matched static vacuum branch,

$$\psi = 0, \qquad T_{\mu\nu} = 0, \qquad \theta = 0, \tag{18}$$

one has

$$V_{\rm quad} = 0, \qquad \frac{dV_{\rm quad}}{d\psi} = 0, \qquad J^\mu_{\rm ops} = 0, \qquad E_\chi = 0, \qquad \mathcal{C}_{\rm min} = 0. \tag{19}$$

So every non-Einstein sector vanishes simultaneously on this exact branch. The action collapses to

$$\boxed{S_{\text{vac}} = \int d^4x \sqrt{-g}\, \frac{R}{16\pi G}.} \tag{20}$$

Therefore the exterior field equation is exactly

$$\boxed{G_{\mu\nu} = 0.} \tag{21}$$

For a static, asymptotically flat, spherically symmetric vacuum exterior, Schwarzschild follows exactly.

This is the strongest exact action-level result currently in hand.

5. Minisuperspace and cosmology after the repair

The repaired potential does not break the previously surviving minisuperspace structure.

5.1. Dust

On the background branch $\psi = 0$,

$$V_{\text{quad}} = 0, \qquad \frac{dV_{\text{quad}}}{d\psi} = 0, \tag{22}$$

and the dust branch remains exact. The count-normalized background equation is

$$\boxed{H^2 = \alpha\rho.} \tag{23}$$

So the $4\pi^2$ dust normalization survives the repair unchanged.

5.2. Radiation

On the branch $\psi = 0$, the repaired potential still has zero value and zero derivative, but the radiation ψ-equation is nonzero. So radiation still sources the lag field away from the background branch. The repaired potential therefore preserves the existence of the nonzero radiation lag branch rather than killing it.

5.3. Vacuum

On the background branch,

$$\rho = 0, \qquad \psi = 0, \tag{24}$$

one gets exact vacuum decoupling and the minisuperspace equation reduces to

$$H^2 = 0. \tag{25}$$

So the repaired action preserves vacuum decoupling in both the cosmological and strong-field senses.

6. Repaired static interior branch

We now turn to the repaired static interior branch with

$$\psi(r) = 0, \qquad \chi = \bar{\chi}(t), \qquad T^{\mu}{}_{\nu} = \mathrm{diag}\big(-\rho(r), p(r), p(r), p(r)\big). \tag{26}$$

Take standard static spherical variables,

$$ds^2 = -e^{2\Phi(r)}dt^2 + \frac{dr^2}{A(r)} + r^2 d\Omega^2, \qquad A(r) = 1 - \frac{2m(r)}{r}. \tag{27}$$

On this branch,

$$\theta = 0, \qquad \mathcal{H}_\chi = 0, \qquad V_{\mathrm{quad}} = 0, \qquad E_\chi = \rho(r), \qquad \rho_{\mathrm{ops}} = \frac{\rho(r) + 3p(r)}{h}. \tag{28}$$

6.1. Exact mass equation

The exact tt-side branch load is

$$\rho_{\mathrm{eff}}^{(tt)} = (1 + \Lambda_C \alpha)\rho(r), \tag{29}$$

so the exact mass equation is

$$\boxed{m'(r) = 4\pi r^2 (1 + \Lambda_C \alpha)\rho(r).} \tag{30}$$

This equation is exact on the repaired $\psi = 0$ branch under the present effective-fluid reading.

6.2. Exact source sign on the radial side

The repaired $\psi = 0$ branch source integrand yields

$$T^{t}{}_{t\,(\mathrm{src})} = 0, \tag{31}$$

and

$$\boxed{T^{r}{}_{r\,(\mathrm{src})} = -\frac{\eta\kappa}{h} e^{-\Phi(r)}\big(\rho(r) + 3p(r)\big).} \tag{32}$$

So the source sector contributes a negative radial-pressure correction and does not enter the mass side directly.

6.3. Cleaned-up modified TOV skeleton

Using perfect-fluid closure for the matter+count part, the effective radial pressure is

$$\boxed{p_{\mathrm{eff}}^{(r)} = (1 + \Lambda_C \alpha)p(r) - \frac{\eta\kappa}{h} e^{-\Phi(r)}\big(\rho(r) + 3p(r)\big).} \tag{33}$$

Therefore the cleaned-up lapse equation is

$$\boxed{\Phi'(r) = \frac{m(r) + 4\pi r^3 \left[(1+\Lambda_C\alpha)p(r) - \frac{\eta\kappa}{h} e^{-\Phi(r)}\big(\rho(r)+3p(r)\big)\right]}{r\,[r-2m(r)]}.} \tag{34}$$

Assuming separate matter conservation on the branch,

$$\boxed{p'(r) = -(\rho(r)+p(r))\Phi'(r).} \tag{35}$$

So the repaired interior branch is described by

$$\boxed{\begin{aligned} m'(r) &= 4\pi r^2(1+\Lambda_C\alpha)\rho(r), \\ \Phi'(r) &= \frac{m(r) + 4\pi r^3 \left[(1+\Lambda_C\alpha)p(r) - \frac{\eta\kappa}{h} e^{-\Phi(r)}\big(\rho(r)+3p(r)\big)\right]}{r\,[r-2m(r)]}, \\ p'(r) &= -(\rho(r)+p(r))\Phi'(r). \end{aligned}} \tag{36}$$

This is the current live modified-TOV branch.

7. Center regularity

The repaired modified-TOV branch is center-regular. Expanding near $r = 0$ as

$$\rho(r) = \rho_c + \rho_2 r^2 + \cdots, \quad p(r) = p_c + p_2 r^2 + \cdots, \quad \Phi(r) = \Phi_c + \Phi_2 r^2 + \cdots, \quad m(r) = m_3 r^3 + m_5 r^5 + \cdots, \tag{37}$$

one finds

$$\boxed{m_3 = \frac{4\pi}{3}(1+\Lambda_C\alpha)\rho_c, \qquad m_5 = \frac{4\pi}{5}(1+\Lambda_C\alpha)\rho_2.} \tag{38}$$

No $1/r$ or constant singular terms appear in the Φ- or p-equations. The branch has

$$\Phi'(r) \sim r, \qquad p'(r) \sim r, \tag{39}$$

so the stellar center is regular on this branch.

8. Weak-field / Newtonian limit

In the weak-field regime,

$$\frac{2m}{r} \ll 1, \qquad |\Phi| \ll 1, \qquad p \ll \rho, \qquad e^{-\Phi} \approx 1, \tag{40}$$

the repaired branch becomes

$$\boxed{m'(r) \approx 4\pi r^2(1+\Lambda_C\alpha)\rho(r),} \tag{41}$$

$$\boxed{\Phi'(r) \approx \frac{m(r)}{r^2} - 4\pi\Gamma r\rho(r),} \qquad \Gamma \equiv \frac{\eta\kappa}{h}, \tag{42}$$

and therefore

$$\boxed{p'(r) \approx -\rho(r)\left[\frac{m(r)}{r^2} - 4\pi\Gamma r\rho(r)\right].} \tag{43}$$

So the Newtonian face of the repaired interior branch is straightforward:

- the count sector rescales the enclosed mass by $(1 + \Lambda_C\alpha)$,
- the source sector adds the local correction $-4\pi\Gamma r\rho$ to the gravitational acceleration.

This is a clean and interpretable weak-field limit.

9. Status ledger

Exact

- repaired potential $V_{\text{quad}} = V_0(e^{-\psi} - 1)^2$,
- exact static vacuum exterior branch collapse,
- exact Schwarzschild recovery in the exterior,
- exact mass equation on the repaired $\psi = 0$ interior branch.

Branch-level / effective-fluid closed

- source-sector structure,
- minimal count-sector structure,
- $\varepsilon = 0$ branch viability,
- dust/radiation/vacuum minisuperspace branches after repair,
- cleaned-up modified-TOV skeleton,
- center regularity,
- weak-field/Newtonian limit.

Still open

- final covariant/background-independent packaging of $\bar{\chi}$,
- full perfect-fluid variational closure for the radial matter/count sector,
- deeper interior phenomenology: regular families, matching, mass-radius relations, and stability,
- full perturbative normalization closure from the repaired action.

10. How this action should be described

The strongest honest description is:

$$\boxed{\text{the effective covariant action for local lag dynamics about the cosmological bookkeeping branch.}} \tag{44}$$

That wording does three important things at once:

- it preserves the one-field logic of the theory,
- it explains why the repaired potential is background-relative rather than absolute,
- and it avoids claiming a fully background-independent microscopic closure before that work is actually done.

A. Appendix: why the absolute potential was replaced

The decisive obstruction was simple. On a static Killing-aligned clock branch, the absolute potential survives as

$$V_{\text{abs}} = V_0 e^{-\chi_0 - \kappa t}, \tag{45}$$

so even though the source and count kinematic pieces vanish in vacuum, the potential and its derivative do not. The repaired quadratic background-relative potential is the first candidate that satisfies both

$$V_{\text{quad}}\big|_{\psi=0} = 0, \qquad \frac{dV_{\text{quad}}}{d\psi}\Big|_{\psi=0} = 0, \tag{46}$$

on the exact static vacuum branch.

B. Appendix: exact versus effective-fluid closure

Two levels of closure are present in the current action program.

1. **Exact branch results.** These include the repaired potential audit, exact static vacuum collapse, exact exterior Schwarzschild recovery, and the exact interior mass equation on the repaired $\psi = 0$ branch.
2. **Effective-fluid branch closures.** These include the cleaned-up radial pressure, the modified-TOV skeleton, center regularity, and the Newtonian limit. These use a standard perfect-fluid closure for the matter/count sector while retaining the exact source correction derived from the reduced branch action.

This division is not a weakness; it is the correct honesty line for the current state of the action.